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Image pickup apparatus

Abstract

An image pickup apparatus includes an image sensor configured to capture an object image, an electronic device configured to perform processing relating to imaging, a duct thermally connected to the electronic device, a first airflow generating unit configured to generate first airflow for cooling the electronic device in the duct, and a second airflow generating unit disposed outside the duct and configured to generate second airflow for cooling the image sensor.

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Background/Summary

BACKGROUND

Technical Field

(1) One of the aspects of the disclosure relates to an image pickup apparatus that internally generates an airflow for cooling.

Description of the Related Art

(2) Image pickup apparatuses such as digital still cameras and video cameras include image sensors such as CMOS sensors and CCD sensors, and electronic devices such as CPUs and ICs mounted on circuit boards, and these components generate heat. In a case where the temperatures of the image sensor and electronic device excessively rise, their performance may deteriorate or malfunction may occur, resulting in poor imaging.

(3) Japanese Patent Laid-Open No. (“JP”) 2020-57890 discloses an image pickup apparatus including an image sensor, a substrate mounted with a plurality of ICs, and a plurality of cooling fans disposed in a duct thermally connected to the substrate.

(4) However, in the image pickup apparatus disclosed in JP 2020-57890, the air generated by the plurality of cooling fans and flowing through the duct mainly cools the substrate, and may not sufficiently cool the image sensor.

SUMMARY

(5) The disclosure provides an image pickup apparatus that can sufficiently cool not only an electronic device but also an image sensor.

(6) An image pickup apparatus according to one aspect of the disclosure includes an image sensor

configured to capture an object image, an electronic device configured to perform processing relating to imaging, a duct thermally connected to the electronic device, a first airflow generating unit configured to generate first airflow for cooling the electronic device in the duct, and a second airflow generating unit disposed outside the duct and configured to generate second airflow for cooling the image sensor.

(7) Further features of the disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. 1A and 1B are perspective views of a digital camera according to a first embodiment.
- (2) FIG. 2 is an exploded perspective view of the digital camera according to the first embodiment.
- (3) FIGS. 3A and 3B are perspective views of a duct in the digital camera according to the first embodiment.
- (4) FIG. 4 is a sectional view of the duct according to the first embodiment.
- (5) FIG. 5 illustrates the duct and a heat source according to the first embodiment.
- (6) FIGS. 6A and 6B are exploded perspective views of an imaging unit according to the first embodiment.
- (7) FIGS. 7A to 7C are rear views of the imaging unit and an internal fan according to the first embodiment.
- (8) FIG. 8 is a block diagram illustrating the configuration of the digital camera according to the first embodiment.

DESCRIPTION OF THE EMBODIMENTS

(9) Referring now to the accompanying drawings, a description will be given of embodiments according to the disclosure.

First Embodiment

(10) FIG. 8 illustrates a configuration of a digital camera (simply referred to as a camera hereinafter) **100** as an image pickup apparatus according to a first embodiment of the disclosure. A lens unit **500** including a lens **501** is attachable to and detachable from (interchangeable from) the camera **100**. Although only one lens **501** is illustrated in FIG. 8 for simplification, a plurality of lenses are actually provided.

(11) In the camera **100**, a shutter **410** is a focal plane shutter that controls the exposure time of an image sensor **161**, which will be described below, and its operation is controlled by a camera control unit **420**, which will be described below. The image sensor **161** includes a CCD sensor or a CMOS sensor, photoelectrically converts (images) an object image (optical image) formed by light passing through the lens **501**, and outputs an imaging signal (analog signal). An A/D converter **412** converts an analog imaging signal output from the image sensor **161** into a digital imaging signal. The digital imaging signal is written into a memory **421** via an image processing unit **413** and a memory control unit (mem ctrl) **422**, which will be described below, or only via the memory control unit **422**.

(12) The image processing unit (IPU) **413** performs image processing such as pixel interpolation processing, resizing processing, and color conversion processing for the digital imaging signal from the A/D converter **412** or the memory control unit **422**, and generates image data. The image processing unit **413** also performs auto white balance (AWB) processing based on a calculation result using the image data.

(13) The camera control unit **420** includes a computer including a processor such as a CPU and circuits, and controls the entire camera **100** and lens unit **500** by executing programs stored in a nonvolatile memory (nv mem) **423**. For example, the camera control unit **420** controls the image

sensor **161** and the shutter **410** according to an imaging instruction by the user, and performs autofocus (AF) control and aperture control based on the image data generated by the image processing unit **413**.

(14) The memory **421** temporarily stores the digital imaging signal output from the A/D converter **412** and the image data generated by the image processing unit **413**. The memory control unit **422** controls data communications among the A/D converter **412**, the image processing unit **413**, and the memory **421**. The nonvolatile memory **423** is an electrically erasable/recordable read-only memory, and stores constants, programs, and the like for the operations of the camera control unit **420**. A system (syst) memory **424** is a readable and writable memory that stores constants and variables for operations of the camera control unit **420**, programs read from the nonvolatile memory **423**, and the like.

(15) A system (syst) timer **425** measures the non-operation time until auto power-off, which puts the camera **100** into a power-saving state to prevent battery consumption in a case where the camera **100** is not operated by the user, and the exposure time of the image sensor **161** by the shutter **410**.

(16) A power supply unit (psu) **430** includes a primary battery, a secondary battery, or an AC adapter. A power supply control unit (pscu) **431** determines whether or not a battery is attached to the power supply unit **430**, determines the type of the attached battery, detects the remaining battery level, and supplies the required voltage to the supply destination at the required timing.

(17) A camera communication terminal **440** is electrically connected to a lens communication terminal **506** provided in the lens unit **500** and enables communication between the camera control unit **420** and a lens control (ctrl) unit **505** within the lens unit **500**.

(18) A recording medium interface (rec med I/F) **441** is an interface with a recording medium (rec med) **600** attachable to and detachable from the camera **100**. The recording medium **600** is a memory card, FLASH memory, hard disk drive, or the like, and records image data (still image and moving image) generated by the image processing unit **413**.

(19) A shake detecting (dtct) unit **442** includes a gyro sensor or the like, and outputs a signal corresponding to the shake of the camera **100** due to manual (camera) shake or the like (referred to as camera shake hereinafter).

(20) The A/D converter **412**, the image processing unit **413**, the camera control unit **420**, the memory **421**, the memory control unit **422**, the nonvolatile memory **423**, the system memory **424**, the system timer **425**, and the power supply control unit **431** described above include a plurality of electronic devices such as a CPU, an IC, memory chips, and are mounted on a main board (substrate) **121**. The recording medium I/F **441** and the shake detecting unit **442** are also mounted on the main board **121**.

(21) The memory **421** described above also serves as a memory for image display (video memory). The digital imaging signal and image data written in the memory **421** are displayed as a live-view image and an image for a confirmation of imaging via the memory control unit **422** on a rear display unit (rear du) **450** provided on the back surface of the camera **100** and an Electric View Finder (EVF) display unit **451** disposed in a viewfinder. The rear display unit **450** and the EVF display unit **451** include display elements such as liquid crystal panels and organic EL panels.

(22) An operation unit **460** is an input unit that accepts an operation by the user, and outputs a signal corresponding to the operation accepted here to the camera control unit **420**. The operation unit **460** includes various operation members such as a mode switch (sw) **461**, a first shutter switch **463**, and a second shutter switch **464** interlocked with a shutter button (btn) **462**, a touch panel **465**, a power switch (psw) **466**, and the like. The mode switch **461** is operated to switch between an imaging mode for still image capturing and moving image capturing. The shutter button **462** is operated by the user to give an imaging preparation instruction and an imaging instruction. The first shutter switch **463** is turned on by half-pressing the shutter button **462** and outputs an SW1 signal to the camera control unit **420**. The second shutter switch **464** is turned on by fully pressing

the shutter button **462** and outputs an SW2 signal to the camera control unit **420**. The camera control unit **420** executes the imaging preparation operation (AF, auto-exposure (AE), AWB, etc.) according to the SW1 signal, and executes still image imaging processing for recording according to the SW2 signal.

(23) The operation unit **460** also includes the touch panel **465** provided on the rear display unit **450**. A power switch **466** is operated to power on and off the camera **100**.

(24) A duct fan (first airflow generating unit) **123** generates airflow in a duct, which will be described below and is thermally connected to the main board **121**, and is provided to cool electronic devices that perform control and processing related to imaging, such as CPUs and ICs as heat sources among many electronic devices mounted on the main board **121**. An internal fan (int fan: second airflow generating unit) **124** is provided to cool the image sensor **161** by generating airflow near the imaging unit **120** including the image sensor **161** as a heat source outside the duct. The camera control unit (control unit) **420** controls the driving (rotation, stop, and the number of rotations) of each of the duct fan **123** and the internal fan **124**. The cooling effect of the duct fan **123** and internal fan **124** will be described below.

(25) In the lens unit **500**, a lens communication terminal **506** is a communication terminal for the lens unit **500** to communicate with the digital camera **100** via the camera communication terminal **440**.

(26) In the lens unit **500**, the lens control unit **505**, which has received a control instruction from the camera control unit **420** by communication, performs position (F-number of aperture value) control of an aperture stop (diaphragm) **503** via an aperture driving (dr) unit **504** and focus control of the lens **501** via a lens driving (dr) unit **502**.

(27) FIGS. **1A** and **1B** illustrate the appearance of the camera **100**. In the following description, an object side of the camera **100** is also called a front side, and a back surface side is also called a rear side. FIG. **1A** illustrates the appearance of the camera **100** viewed obliquely from the front side, and FIG. **1B** illustrates the appearance of the camera **100** viewed obliquely from the rear side. **0** denotes an imaging optical axis from the lens unit **500** to the image sensor **161**.

(28) A plurality of operating members that can be operated by the user are attached to a rear cover **101**. The viewfinder unit that includes the EVF display unit **451** and an eyepiece unit **443** to which a user observing the EVF display unit **451** brings his eye **700** closer as illustrated in FIG. **8** is attached to the rear cover **101**.

(29) A duct, which will be described below, is provided inside the camera **100** (between the main board **121** and the rear cover **101**), and the rear cover **101** is formed with ventilation ports **110a** that allow air to enter the duct. A display housing unit **102** is provided to house the rear display unit **450** as a variable angle monitor that is rotatable relative to the camera **100**. A recording medium lid **103** is a cover that closes an opening of a slot that houses the recording medium (memory card) **600** described above.

(30) A mount **112** to which the lens unit **500** is attached is fixed to a front base **104**, and a grip unit **105** for gripping the camera **100** by the user is provided to the front base **104**. A battery lid **108a** is attached to a bottom cover **108** to close an opening of a battery chamber. The bottom cover **108** also has an opening for exposing a tripod seat **106** fixed to a bottom surface portion of the front base **104**.

(31) A top cover **107** is provided with a plurality of operating members (mode switch **461**, shutter button **462**, power switch **466**, etc.) that can be operated by the user. A terminal cover **111** for protecting an external communication terminal **121c**, which will be described below, is attached to a side cover **109**, and ventilation ports **110b** for exhausting air from the duct are formed in the side cover **109**.

(32) FIG. **2** is an exploded view of the camera **100** illustrated in FIGS. **1A** and **1B**. The rear cover **101**, the front base **104**, the top cover **107**, the bottom cover **108**, and the side cover **109** described above form the exterior of the camera **100**. An imaging unit **120**, the main board **121**, the duct **122**,

the shutter **410**, the EVF display unit **451**, and the like are provided Inside the camera **100**.

(33) The main board **121** is mounted with a plurality of electronic devices (referred to as a heat-generating element group hereinafter) **121a** that serve as heat generating sources, a recording medium I/F **121b** (**441** in FIG. **8**), and the external communication terminal **121c** to which a cable to be connected to an unillustrated external device.

(34) The duct **122** is formed so that air from the outside of the camera **100** is drawn into an area near the heat-generating element group **121a**. The duct fan **123** is disposed inside the duct **122**.

(35) Since the heat-generating element group **121a** consumes a particularly large amount of power and generates a large heat value (calorific value) in the camera **100**, the imageable time of the camera **100** is limited by the guaranteed operating temperature of the heat-generating element group **121a**, excluding the remaining battery power. In order to maximize the imageable, it is necessary to cool the heat-generating element group **121a** so that their temperature does not exceed the guaranteed operating temperature. This embodiment provides the duct **122** on the rear side of the main board **121** as illustrated in FIG. **2** to thermally connect the heat-generating element group **121a** and the duct **122** to each other, to generate the airflow (first airflow) in the duct **122** by the duct fan **123**, and to cool the heat-generating element group **121a**.

(36) The internal fan **124** is disposed around the imaging unit **120** disposed on the front side of the main board **121** and generates an airflow (second airflow) that flows along the back surface of the image sensor **161** opposite to the imaging plane to cool the image sensor **161**. In this embodiment, the internal fan **124** is disposed so that the air discharge direction is orthogonal to the imaging optical axis O of the camera **100**. However, as long as the air discharged from the internal fan **124** is applied to the image sensor **161**, the discharge direction does not have to be orthogonal to the imaging optical axis O.

(37) Disposing the outlet (discharge port) of the internal fan **124** so as to flow air between the imaging unit **120** and the main board **121** can cool both the image sensor **161** and the heat-generating element group **121a**. Since the imaging unit **120** in this embodiment has a movable portion **160**, which will be described below, and a heat radiation path is narrow, disposing the outlet of the internal fan **124** at a position closer to the imaging unit **120** than the main board **121** can more effectively cool the image sensor **161**.

(38) This embodiment uses a centrifugal fan for the duct fan **123** and the internal fan **124** as the airflow generating units. However, other fans such as an axial (flow) fan may be used. A blower or the like may be used as an airflow generating unit other than a fan.

(39) A detailed description will be given of the duct **122**. FIG. **3A** illustrates the duct **122** viewed obliquely from the front, and FIG. **3B** illustrates the duct **122** viewed obliquely from the rear. FIG. **4** illustrates a section of the duct **122** illustrated in FIG. **3B** taken along a line A-A.

(40) The duct **122** includes a first duct portion **130**, a second duct portion **131**, and a third duct portion **132**. The first duct portion **130** forms a duct extending in the longitudinal (front to rear) direction parallel to the imaging optical axis O between the display housing unit **102** and the EVF display unit **451**. Inlet ports **135** is formed in a rear end surface of the first duct portion **130**.

(41) A top portion of the second duct portion **131** is connected to a front portion of the first duct portion **130** to form a duct extending in a vertical direction orthogonal to the imaging optical axis O. Disposing the first duct portion **130** near the EVF display unit **451** and connecting it to the top portion of the second duct portion **131** can secure a wide thermal connection area between the second duct portion **131** and the heat-generating element group **121a**. This structure is advantageous in terms of heat radiation from the heat-generating element group **121a**.

(42) The second duct portion **131** is made by combining a second front duct portion **131a** forming a front surface on the side of the heat-generating element group **121a** and a second rear duct portion **131b** forming a rear surface on the opposite side of the front surface. A second duct concave portion **131c** is formed in an area of the front surface of the second duct portion **131** that does not face the heat-generating element group **121a** (that is not thermally connected to the heat-generating

element group **121a** via a heat transfer member, which will be described below). Forming the second duct concave portion **131c** locally reduces the sectional area inside the second duct portion **131**. As a result, a large airflow amount inside the area of the second duct portion **131** that is thermally connected to the heat-generating element group **121a** through the heat transfer member in the second duct portion **131** can be secured and efficiently radiate heat from the heat-generating element group **121a**.

(43) In addition, the heat in the camera **100** can be efficiently diffused in the area of the second duct portion **131** that does not face the heat-generating element group **121a**. Therefore, providing the second duct concave portion **131c** in the area that does not face the heat-generating element group **121a** in the second duct portion **131** can achieve both efficient heat radiation from the heat-generating element group **121a** and efficient heat diffusion in the camera **100**.

(44) A connector and flexible printed circuits (FPC) for connecting the imaging unit **120** and the main board **121** to each other are disposed in a space on the side of the main board **121** created by the second duct concave portion **131c**.

(45) A bottom portion of the second duct portion **131** is connected to a rear portion of the third duct portion **132**. The third duct portion **132** is formed so that it extends in the longitudinal direction in parallel with the imaging optical axis O under the imaging unit **120** and the main board **121**. An opening is formed in a rear area of the bottom surface of the third duct portion **132**, and the duct fan **123** is disposed in the third duct portion **132** through this opening. The opening is closed by fixing a bottom-surface duct cover **133** to the third duct portion **132** with a screw **137** via an unillustrated elastic member. At this time, the duct fan **123** is also fixed to the third duct portion **132** with the screw **137**. An exhaust port **136** is formed in the rear end surface of the third duct portion **132**.

(46) A tripod seat mount portion **140** having a concave shape is formed in a front area of the bottom surface of the third duct portion **132** where the exhaust port **136** is not formed. Part of the tripod seat **106** fixed to the front base **104** is accommodated in the tripod seat mount portion **140**. The front base **104** is made of a magnesium alloy so as to have high rigidity. Thereby, deformation of the camera **100** fixed to a tripod can be suppressed even in a case where the camera **100** receives a strong external force, and as a result, deformation of the duct **122** can be also suppressed.

(47) The inlet ports **135** of the thus-configured duct **122** (the first duct portion **130**) are connected to the ventilation ports **110a** of the rear cover **101**. At this time, an elastic member is interposed between the periphery of the inlet ports **135** in the duct **122** and the periphery of the ventilation ports **110a** in the rear cover **101** for sealing, and thereby air is prevented from flowing into the inlet ports **135** from a gap other than the ventilation ports **110a**.

(48) The exhaust port **136** of the duct **122** (the third duct portion **132**) is connected to the ventilation ports **110b** of the side cover **109**. At this time, an elastic member is interposed between the periphery of the exhaust port **136** in the duct **122** and the periphery of the ventilation ports **110b** in the side cover **109** for sealing, and thereby the air is prevented from flowing out from a gap other than the ventilation ports **110b**.

(49) In order to reduce the pressure loss in the duct **122**, the flow passages of the first duct portion **130**, the second duct portion **131**, and the third duct portion **132** may have sectional areas as equal as possible. Even in a case where the flow passages have different sectional areas, each duct portion may be configured so that the difference in sectional area becomes as small as possible.

(50) The first duct portion **130**, the second duct portion **131**, and the third duct portion **132** may be formed as separate parts and connected to one another by screwing, bonding, crimping, or the like, or integrally formed by a 3D printer or the like. In the case where they are connected to one another, the airtightness of the duct **122** may be secured by inserting the elastic member therebetween.

(51) Each of the first to third duct portions **130** to **132** may be made of a material with high thermal conductivity so that heat from the heat source can be efficiently transferred. Since the contribution

of the second rear duct portion **131b** to heat diffusion from the heat source is relatively small, it may be made of a material having a lower thermal conductivity than the second front duct portion **131a**. From the viewpoint of weight reduction of the camera **100**, each of the first to third duct portions **130** to **132** may be made of a material with low specific gravity. If the main material of the duct **122** is metal, it can be used as a ground inside the camera **100**, which is advantageous from the standpoint of noise reduction. In this embodiment, the first duct portion **130**, the second front duct portion **131a**, and the third duct portion **132** is made of aluminum, and the second rear duct portion **131b** is made of polycarbonate.

(52) Integrating the second rear duct portion **131b** with the rear cover **101** can reduce the thickness of the camera **100** in the longitudinal direction.

(53) In this embodiment, a plurality of first fin portions **141** are formed inside the second front duct portion **131a**, and a plurality of second fin portions **142** are formed inside the third duct portion **132**. Thereby, the surface area in the second duct portion **131** and the third duct portion **132**, that is, the heat exchange area with the air can be increased to enhance the cooling efficiency.

(54) In a case where the duct fan **123** is driven, outside air that has passed through the ventilation ports **110a** flows into the duct **122** through the inlet ports **135**. The air flowing into the duct **122** flows along the first fin portions **141** in the second duct portion **131** and the second fin portions **142** in the third duct portion **132** to which the heat from the heat-generating element group **121a** is transferred, and receives heat from these fin portions **141** and **142**. The heated air is sucked from the inlet port of the duct fan **123**, is discharged from the discharge port, and is discharged to the outside of the camera **100** through the exhaust port **136** and the ventilation ports **110**. In this embodiment, the air is exhausted from the ventilation ports **110b** formed in the side cover **109**. This is to prevent the exhaust air from directly hitting the face of the user. However, the ventilation port for exhaust may be formed at another location.

(55) FIG. 5 illustrates a thermal connection structure between the duct **122** and the heat source. FIG. 5 illustrates the second front duct portion **131a**, the third duct portion **132**, the main board **121**, and the plurality of heat transfer members **150** in a case where the duct **122** illustrated in FIG. 3B is taken along a line C-C. Each heat transfer member **150** is a heat conductive rubber for thermally connecting the heat-generating element group **121a** and the second front duct portion **131a** to each other, and contacts the front surfaces of the heat-generating element group **121a** and the second front duct portion **131a**. The heat generated in the heat-generating element group **121a** is transmitted to the second front duct portion **131a** (that is, the duct **122**) via the heat transfer members **150**. The first fin portions **141** are provided in the (rear) area of the second front duct portion **131a** which the heat transfer members **150** contact. The heat transmitted from the heat-generating element group **121a** to the first fin portions **141** is transmitted to the air flowing along the first fin portions **141** in the second duct portion **131**, and the warmed air is exhausted from the exhaust port **136** through the third duct portion **132**. Thus, heat exchange is efficiently performed, and the heat-generating element group **121a** is sufficiently cooled.

(56) The heat transfer member **150** is not limited to heat conductive rubber, and may be a metal foil, a heat pipe, or the like, as long as it has a higher thermal conductivity than a resin member and satisfies the required mechanical and electrical properties.

(57) FIGS. 6A and 6B illustrate the imaging unit **120** in an exploded manner from the oblique front side and the oblique rear side, respectively. The imaging unit **120** has the movable portion **160** and a drive base **164**. The movable portion **160** includes the image sensor **161** and a sensor holder **163** that holds the image sensor **161**. The image sensor **161** is made by bonding a sensor chip having a plurality of pixels to an imaging substrate **166** and electrically connecting electrodes of the sensor chip and an imaging circuit on the imaging substrate **166** by wire bonding. The image sensor **161** is fixed to the sensor holder **163** by adhesion. Electronic devices **166a** such as capacitors, resistors, and regulators, that constitute the imaging circuit are mounted on a back surface of a sensor-chip adhesion surface of the imaging substrate **166**.

(58) The sensor holder **163** is held by the drive base **164** so as to be movable in two mutually orthogonal directions (horizontal direction and vertical direction) orthogonal to the imaging optical axis O. Three coils **162** are fixed to the sensor holder **163**. The drive base **164** holds three magnets **165** so as to face the three coils **162**. The movable portion **160** is attracted in the imaging optical axis direction (rear side) by the magnetic force of the magnet **165**. Unillustrated balls held by ball holding portions **163a** provided at a plurality of locations of the sensor holder **163** are disposed between the movable portion **160** and the drive base **164**. Thereby, the movable portion **160** is positioned in the imaging optical axis direction relative to the drive base **164** via the balls.

(59) In the thus-configured imaging unit **120**, the image sensor **161** can be moved in the above two directions by controlling the energization of the three coils **162**. The camera control unit **420** controls the energization of the coils **162** according to the camera shake detected by the shake detecting unit **442** so as to move the movable portion **160** in a direction to reduce (correct) the image blur caused by the camera shake.

(60) The imaging unit **120** and the main board **121** are electrically connected to each other via an FPC. An imaging signal FPC **167** has wirings for transmitting an imaging signal output from the image sensor **161** and a control signal necessary to drive the image sensor **161**, and these signals are sent to the camera control unit **420** on the main board **121**. An imaging power supply FPC **168** has wirings for supplying power for driving the image sensor **161** from the power supply control unit **431** to the image sensor **161**.

(61) Referring now to FIGS. 7A to 7C, a description will be given of a positional relationship between the movable portion **160** that moves in the imaging unit **120** and the internal fan **124**. FIG. 7A illustrates the internal fan **124** and the imaging unit **120** viewed from the rear, and FIG. 7B schematically illustrates them. FIG. 7C schematically illustrates how the air discharged from the internal fan **124** flows between the imaging unit **120** and the main board **121**.

(62) As illustrated in FIG. 7C, an orientation of a discharge port **124a** of the internal fan **124** is set so that air flows between the imaging unit **120** and the main board **121**. Arrows in FIG. 7C indicates a flow of air sucked into the internal fan **124** and air discharged from the internal fan **124**.

(63) In FIG. 7B, reference numeral **160a** denotes a moving range of the movable portion **160** in the direction orthogonal to the imaging optical axis, and reference numeral **160b** denotes a range in which the image sensor **161** exists (predetermined range: referred to as sensor moving range hereinafter) even if the movable portion **160** moves to a position within the moving range **160a**. In a case where a range L in which the air discharged from the discharge port **124a** of the internal fan **124** flows when it is viewed from the imaging optical axis direction overlaps the sensor moving range **160b**, even if the image sensor **161** moves to any positions within the sensor moving range **160b**, the air discharged from the fan **124** flows along the back surface of the image sensor **161** and can cool the image sensor **161**. The internal fan **124** that is not physically connected to the movable portion **160** does not hinder the movement of the movable portion **160** (that is, the image stabilizing function).

(64) As described above, in this embodiment, in addition to the duct fan **123** disposed in the duct **122** for cooling the heat-generating element group **121a** on the main board **121**, the internal fan **124** configured mainly to cool the image sensor **161** is disposed inside the camera **100**. Thereby, temperature rises of the heat-generating element group **121a** and the image sensor **161** can be suppressed, and malfunction of the camera **100** caused by the excessive temperature rise can be suppressed. In particular, in the case where the image sensor **161** is movable as in this embodiment, the image sensor **161** cannot be thermally connected to the duct **122** using a heat transfer member unlike the heat-generating element group **121a**. Therefore, providing the internal fan **124** can suppress the temperature rise of the movable image sensor **161**.

(65) A temperature analysis results using a finite element method (FEM) will be illustrated. This embodiment can suppress the temperatures of the heat-generating element group **121a** and the image sensor **161** by 6° C. or lower in comparison with a case where neither duct fan **123** nor the

internal fan **124** is provided.

(66) Referring now to Table 1, a description will be given of control modes of the duct fan **123** and the internal fan **124** in the camera **100**. Table 1 illustrates the number of rotations (NoR) of the duct fan **123** and the internal fan **124** in each control (ctrl) mode, and an imaging mode (imaging condition) suitable for each control mode. In each control mode, the rotations of the duct fan **123** and the internal fan **124** are controlled so that the number of rotations of the duct fan **123** and the number of rotations of the internal fan **124** have a predetermined relationship.

(67) TABLE-US-00001

TABLE 1	NoR DUCT FAN	INT FAN	CTRL	NoR	NoR	NoR	NoR	NoR
NoR IMAGING MODE	MODE (25)	(50)	(100)	(25)	(50)	(100)	IMAGE QUALITY	SP/CP 1ST
○ FULL HD C 2ND	○	○	4K C 3RD	○	○	8K C 4TH	○	○ FULL HD S 5TH
○	○	4K S 6TH	○	○	8K S 7TH	○	(○)	(○)
(○)	(○)	(○)	(○)	(○)	(○)	(○)	(○)	STILL IMAGE/PWR-OFF C

(68) Both the duct fan **123** and the internal fan **124** are controlled by the camera control unit **420** so as to rotate at three different numbers of rotations: the number of rotations (25), the number of rotations (50), and the number of rotations (100). The number of rotations becomes higher in the order of the number of rotations (25)<the number of rotations (50)<the number of rotations (100). A parenthesized figure attached to the number of rotations is a merely numerical value for comparing the height of the number of rotations, and does not represent an actual number of rotations or a ratio of the number of rotations.

(69) The camera control unit **420** controls the number of rotations of the duct fan **123** and the number of rotations of the internal fan **124** in first to seventh control modes according to the heat values generated by the image sensor **161** and the main board **121** (heat-generating element group **121a**).

(70) A camera including a fan has a risk that the rotation sound of the fan is recorded on a recording medium as noise if it is transmitted to a built-in microphone. Hence, the camera **100** in which the duct fan **123** is disposed in the duct **122** having the inlet ports **135** and the exhaust port **136** as in this embodiment particularly needs to suppress the number of rotations of the duct fan **123** as low as possible to reduce that noise. Of the three numbers of rotations, the number of rotations (25) is the lowest, and noise is less likely to be recorded.

(71) On the other hand, since the internal fan **124** is disposed outside the duct **122** inside the camera **100**, its rotation sound is less likely to be transmitted to the microphone than that of the duct fan **123**. Therefore, the number of rotations of the internal fan **124** can be set with priority given to cooling (C) the image sensor **161** over noise reduction.

(72) The camera **100** according to this embodiment can record moving images with three types of image quality (number of imaging pixels) of Full HD, 4K, and 8K. The higher the image quality of the moving image is, the higher the power consumption becomes for the calculation processing and recording of the image data. Thus, the heat values of the image sensor **161** and the main board **121** increase. Moreover, in still image capturing, power consumption for the calculation processing and recording of the image data is less than that in moving image capturing and the heat value generated from the combination of the image sensor **161** and the main board **121** becomes lower. Therefore, the heat value generated from the combination of the image sensor **161** and the main board **121** becomes higher in the order of still image capturing<Full HD moving image capturing<4K moving image capturing<8K moving image capturing. Thus, the heat value generated from the combination of the image sensor **161** and the main board **121** is different according to the imaging mode, so the camera control unit **420** controls the number of rotations of each of the duct fan **123** and the internal fan **124** in a control mode that depends on the imaging mode. The camera control unit **420** sets a control mode according to a still or moving image capturing mode selected by the user through the mode switch **461**, and a detailed imaging mode such as image quality, cooling priority (CP), and silence priority (SP) selected from the menu screen displayed on the rear display unit **450**.

(73) The first to third control modes are control modes that give priority to cooling the image

sensor **161**. In these first to third control modes, the camera control unit **420** sets the number of rotations of the duct fan **123** to the number of rotations (50) regardless of the control mode, while changing the number of rotations of the internal fan **124** for each control mode. The duct fan **123** has a high cooling effect on the main board **121**, but a low cooling effect on the image sensor **161** which is distant from and located on the front side of the main board **121**. Therefore, in the first to third control modes, the number of rotations of the duct fan **123** is fixed to the number of rotations (50) at which noise is not significantly influential.

(74) The camera control unit **420** sets the number of rotations of the internal fan **124** to the lowest number of rotations (25) in the first control mode. In this case, the cooling effect of the internal fan **124** on the image sensor **161** becomes low. Thus, the first control mode is a control mode suitable for Full HD moving image capturing, which generates a small heat value.

(75) The camera control unit **420** sets the number of rotations of the internal fan **124** to the number of rotations (50) in the second control mode. The number of rotations (50) is between the number of rotations (25) and the number of rotations (100), and the cooling effect of the internal fan **124** on the image sensor **161** at this time is standard. Thus, the second control mode is a control mode suitable for standard 4K moving image capturing.

(76) The camera control unit **420** sets the number of rotations of the internal fan **124** to the highest number of rotations (100) in the third control mode. At this time, the cooling effect of the internal fan **124** on the image sensor **161** also increases. The third control mode is a control mode suitable for 8K moving image capturing, which generates a large heat value.

(77) The fourth to sixth control modes are control modes that give priority to silence (S). In these fourth to sixth control modes, the camera control unit **420** sets the number of rotations of the duct fan **123** to the number of rotations (25) that generates less noise regardless of the control mode, while changing the number of rotations of the internal fan **124** for each control mode.

(78) The camera control unit **420** sets the number of rotations of the internal fan **124** to the lowest number of rotations (25) in the fourth control mode. In this case, the cooling effect of the internal fan **124** on the image sensor **161** becomes low. Thus, the fourth control mode is a control mode suitable for Full HD moving image capturing, which generates a small heat value.

(79) The camera control unit **420** sets the number of rotations of the internal fan **124** to the number of rotations (50) in the fifth control mode. The cooling effect of the internal fan **124** on the image sensor **161** at this time is standard. Thus, the fifth control mode is a control mode suitable for standard 4K moving image capturing.

(80) The camera control unit **420** sets the number of rotations of the internal fan **124** to the highest number of rotations (100) in the sixth control mode. At this time, the cooling effect of the internal fan **124** on the image sensor **161** also increases. Thus, the sixth control mode is a control mode suitable for 8K moving image capturing, which generates a large heat value.

(81) The seventh control mode is a control mode that prioritizes cooling of the image sensor **161** and the main board **121** over noise reduction over the first to third control modes. The camera control unit **420** sets the number of rotations of the duct fan **123** to the highest number of rotations (100) in the seventh control mode. On the other hand, the number of rotations of the internal fan **124** is set to one of the number of rotations (25), the number of rotations (50), and the number of rotations (100).

(82) In a case where the number of rotations of the duct fan **123** is set to the number of rotations (100), the cooling effect on the main board **121** is high, but the rotation sound is likely to be recorded as noise. Thus, the seventh control mode is not suitable for moving image capturing but is suitable for cooling the main board **121** in still image capturing in which the microphone does not operate or in a power-off state (in a case where capturing is stopped). Sufficiently cooling the main board **121** in still image capturing or in the power-off state can lower the temperatures of the image sensor **161** and the main board **121** in a case where still image capturing is switched to moving image capturing and the influence of the heat of the image sensor **161** on image quality can be

suppressed.

(83) Regarding the number of rotations of the internal fan **124**, rotating the internal fan **124** at the number of rotations (25) or higher will provide a sufficient cooling effect because the heat value generated by the image sensor **161** is small in still image capturing or in the power-off state.

(84) Thus, controlling the number of rotations of each of the duct fan **123** and the internal fan **124** according to the first to seventh control modes that depend on the imaging mode (still or moving image capturing and image quality) and the power-off state of the camera **100** can satisfactorily cool the main board **121** and the image sensor **161**. In addition, imaging is available in the noise suppressed state.

Second Embodiment

(85) Table 2 illustrates the number of rotations of each of the duct fan **123** and the internal fan **124** in eighth and ninth control modes in the camera **100** according to a second embodiment and the imaging mode suitable for each control mode. In each control mode, the rotations of the duct fan **123** and the internal fan **124** are controlled so that the number of rotations of the duct fan **123** and the number of rotations of the internal fan **124** (including 0, which is the number of rotations in the stops state) have a predetermined relationship.

(86) TABLE-US-00002
TABLE 2 NoR DUCT FAN INT FAN CTRL STOP NoR NoR NoR NoR
IMAGING MODE MODE (0) (25) (25) (50) (100) IMAGE QUALITY SP/CP 8TH ○ ○ STILL
IMAGE/PWR-OFF S 9TH ○ (○) (○) (○) FULL HD/4K/8K S

(87) The eighth control mode, similar to the seventh control mode described in the first embodiment, is a control mode suitable for still image capturing or the power-off state. In this control mode, the internal fan **124** is stopped in a case where a heat value generated by the image sensor **161** is sufficiently small in still image capturing or in the power-off state. On the other hand, the internal fan **124** is rotated at the number of rotations (100) that provides a high cooling effect.

(88) As described in FIGS. 6A to 7C, the internal fan **124** can have a cooling effect not only on the image sensor **161** but also on the main board **121**. In a case where the image sensor **161** generates a small heat value, rotating only the internal fan **124** at the number of rotations (100) can provide a sufficient cooling effect not only for the image sensor **161** but also for the main board **121**.

Suppressing the power consumption of the camera **100** can be thus expected by stopping the duct fan **123** (by setting the number of rotations to 0) even if only the internal fan **124** is rotated at the number of rotations (100).

(89) The ninth control mode is a control mode that gives priority to silence, similar to the fourth to sixth control modes. In this control mode, the duct fan **123** is rotated at the number of rotations (25) that generates less noise. On the other hand, the internal fan **124** is rotated at any one of the number of rotations (25), the number of rotations (50), and the number of rotations (100) according to the image quality for the moving image. For example, the number of rotations (25) is set to Full HD image quality, the number of rotations (50) is set to 4K image quality, and the number of rotations (100) is set to 8K image quality. This configuration provides a sufficient cooling effect on the image sensor **161**.

(90) In this embodiment, the internal fan **124** can cool the image sensor **161** and the main board **121** according to the imaging mode while securing silence by stopping the duct fan **123** or by rotating it at the small number of rotations (25).

(91) Each of the above embodiment can sufficiently cool not only the electronic device but also the image sensor in the image pickup apparatus.

OTHER EMBODIMENTS

(92) Embodiment(s) of the disclosure can also be realized by a computer of a system or apparatus that reads out and executes computer-executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit

(ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer-executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer-executable instructions. The computer-executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read-only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

(93) While the disclosure has been described with reference to exemplary embodiments, it is to be understood that the disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(94) This application claims the benefit of Japanese Patent Application No. 2021-199528, filed on Dec. 8, 2021, which is hereby incorporated by reference herein in its entirety.

Claims

1. An image pickup apparatus comprising: an image sensor configured to capture an object image; an electronic device configured to perform processing relating to imaging; a duct thermally connected to the electronic device; a first airflow generating mechanism configured to generate first airflow for cooling the electronic device in the duct; a second airflow generating mechanism disposed outside the duct and configured to generate second airflow for cooling the image sensor; and at least one processor programmed to perform operations to individually control drives of the first and second airflow generating mechanisms, wherein the at least one processor is programmed to increase a number of rotation of the second airflow generating mechanism as image quality becomes higher, and the at least one processor programmed to rotate the first airflow generating mechanism at a lower number of rotation in a silence priority mode than the number of rotation of the first airflow generating mechanism in a cooling priority mode.
2. The image pickup apparatus according to claim 1, wherein the second airflow generating mechanism is disposed so that the second airflow flows along a back surface of the image sensor opposite to an imaging plane of the image sensor.
3. The image pickup apparatus according to claim 2, wherein the second airflow generating mechanism is disposed so that the second airflow flows between the back surface of the image sensor and the electronic device to cool the electronic device and the image sensor.
4. The image pickup apparatus according to claim 1, wherein the at least one processor individually controls drives of the first and second airflow generating mechanisms according to an imaging condition of the image pickup apparatus.
5. The image pickup apparatus according to claim 4, wherein the imaging condition determines at least one of still image capturing, moving image capturing, the number of imaging pixels, giving priority to cooling, and giving priority to silence.
6. The image pickup apparatus according to claim 1, wherein the at least one processor controls driving of each of the first and second airflow generating units mechanisms during stopping of imaging.
7. The image pickup apparatus according to claim 1, wherein the at least one processor stops the first airflow generating mechanism at least one of during still image capturing and during stopping

of imaging.

8. The image pickup apparatus according to claim 1, wherein the image sensor is movable within a predetermined range relative to an imaging optical axis for image stabilization, and wherein the second airflow generating mechanism is disposed so that the image sensor is cooled by the second airflow regardless of a position of the image sensor within the predetermined range.
